

Resource Allocation and Resilience Analysis of Human-Humanoid Mixed Production Lines under High Load – A Simulation-based Approach

HUNG-WEI CHIU¹

¹ Department of Computer Science, National Taichung University of Education
acs111851@gm.ntcu.edu.tw

1. Abstract

Recent breakthroughs in Vision-Language-Action (VLA) models have accelerated the integration of humanoid robots into manufacturing a cornerstone of the Industry 5.0 paradigm. However, unlike traditional stationary automation, humanoids navigating non-standardized tasks remain highly susceptible to AI cognitive failures and unpredictable edge cases. While the industry heavily anticipates the productivity gains of embodied AI, the systemic impact of high-frequency logical failures on production line stability remains critically under-quantified. Addressing this imperative, this study develops a novel Human-Robot Collaboration (HRC) simulation framework that hybridizes Discrete Event Simulation (DES) and Agent-Based Modeling (ABM). To capture real-world operational dynamics, our model integrates a complexity-driven cognitive failure function for robots alongside the Jaber-Neumann fatigue model, which precisely quantifies human physiological degradation under high-stress conditions. Through comprehensive Monte Carlo scenario analyses, the empirical data reveals a stark operational threshold: over-allocating humanoid robots inevitably triggers maintenance capacity saturation and catastrophic system collapse. Conversely, preserving a critical baseline of human personnel approximately 30% of the workforce functions as an indispensable "resilience buffer," effectively absorbing the stochastic shocks of AI logic failures and preventing cascading deadlocks. By defining these physical safety boundaries, this research provides actionable, risk-aware resource allocation guidelines for the scaled deployment of humanoid robots in next-generation intelligent manufacturing.

Keywords: Human–Robot Collaboration, HRC、Humanoids、Agent–Based Modeling ABM、Edge Cases、Uncertainty Modeling、Systemic Resilience

2. Introduction

Recent breakthroughs in generative artificial intelligence (AI) have significantly enhanced the feasibility of deploying embodied AI and intelligent robots in manufacturing environments. [1] Specifically, the advancement of Vision-Language-Action (VLA) models has facilitated the practical realization of general-purpose humanoid robots (e.g., Figure 01 and Tesla Optimus) [2]. As these humanoids are expected to gradually integrate into traditional, human-centric factory environments, investigating the future landscape of human-robot collaboration (HRC) has become increasingly crucial. While human-machine collaborative production has been ubiquitous since industrialization, conventional automated environments primarily rely on stationary, single-purpose robotic arms integrated with assembly lines. Although these systems are highly quantifiable and ideal for standardized operating procedures (SOPs), they are strictly confined to fixed-range, repetitive tasks. Conversely, while humanoid robots offer remarkable versatility, they introduce inherent uncertainty risks, including AI-specific hallucinations and cognitive or logical failures when encountering edge

cases or exceptions [3].

Given these emerging application scenarios, conventional factory designs are evidently inadequate for accommodating dynamic HRC. To address this gap, this study develops a hybrid simulation model utilizing the AnyLogic platform, integrating Discrete Event Simulation (DES) and Agent-Based Modeling (ABM). This model simulates a future HRC factory environment by configuring mixed production lines comprising both intelligent robots and human personnel. Through what-if scenario testing and sensitivity analysis, this study identifies the "optimal mixture ratio" and the "break-even point" between human workers and humanoid robots concerning production volume and stability. Furthermore, this study explores the feasibility of personnel scheduling, mutual collaboration, and resource allocation. By evaluating the critical value of human workers in risk mitigation owing to their high adaptability and low error rates this study provides a quantitative reference framework for the future integration of humanoid robots into manufacturing facilities. In summary, the main contributions are as follows:

- **Quantification of VLA Cognitive Failures:** This study constructs a stochastic model incorporating VLA failure characteristics to quantify the impact of logical uncertainties on the operational scheduling and management of robots within an HRC factory setting.
- **Development of a Dynamic Collaboration Architecture:** This study establishes a highly parameterized HRC simulation framework to model future robot integration scenarios. This framework investigates the feasibility and operational impacts of various human-robot configuration ratios, providing a dynamic tool capable of adapting to future technological advancements and the maturation of humanoid product lines.
- **Energy Cycle Modeling using Empirical Data:** Based on the empirical battery capacity and fast-charging specifications of state-of-the-art commercial humanoid robots as of 2026, this study quantifies the impact of long-term charging cycles on system scheduling. This provides a reliable baseline for power consumption and operational endurance for current intelligent robot prototypes.

3. Literature Review

3.1 Human Fatigue and Performance Degradation

Literature indicates that fatigue is a primary precursor to performance degradation [3]. As noted in [4], an increase in human fatigue directly and negatively impacts overall system productivity. To accurately reflect quality degradation and productivity loss in human workers, this study introduces a fatigue coefficient to simulate the cumulative fatigue experienced during work shifts. Furthermore, unlike the standardized execution times of robots, human task durations are inherently variable. To account for this natural variance, human processing time is modeled using a triangular distribution, providing a more realistic representation of human operational dynamics.

3.2 The RT-2 (Vision-Language-Action, VLA) Model

This study builds upon the RT-2 model [5] as a foundational reference for robotic behavior. In the context of Vision-Language-Action (VLA) models, "edge cases" typically occur when the model encounters unseen objects, backgrounds, or environments. To parameterize human-robot interactions, **this study introduces** a cognitive failure parameter that synthesizes the success rate under standard conditions with the failure rate in unobserved edge cases, thereby simulating the overall probability of a robot encountering a failure scenario. Based on RT-2 metrics, the success rate for Seen Tasks is highly robust, ranging from 91% to 93%. However, in Unseen Average scenarios—where the robot encounters novel objects or environments—the average success rate drops to approximately 62%. In

Hard Unseen Environments, characterized by extreme visual complexity, the success rate further declines to merely 33% - 35%.

3.3 Locomotion Depletion and Energy Configuration Benchmarks

Reflecting the state-of-the-art in humanoid robotics as of 2026, this study references industry data from institutions such as TrendForce [6-10]. According to the specifications of mainstream humanoid robots released between 2025 and early 2026, bipedal humanoids equipped with VLA or foundational general AI capabilities typically feature battery capacities ranging from 2 kWh to 2.5 kWh. Under various payloads, their continuous operational endurance generally spans 2 to 5 hours. Consequently, this system configures the maximum continuous operational time under full load (BatteryCapacity) for robot agents at an average of 240 minutes (4 hours). To address the realities of continuous factory production and energy bottlenecks, our model simulates fast-charging protocols based on commercial units like the Unitree G1. When depleted, robots are forced to exit the production line for a 90-minute (1.5-hour) autonomous fast-charging cycle. This configuration is designed to accurately simulate the prolonged latency and disruptions that lengthy energy replenishment cycles impose on overall production scheduling.

4. Methodology & Model Construction

To establish a generalized Human-Robot Collaboration (HRC) framework, this study develops a hybrid simulation model integrating Discrete Event Simulation (DES) and Agent-Based Modeling (ABM) via the AnyLogic platform. The core of the system architecture is centered on intervention scenarios specifically, human operators assisting intelligent robots during operational failures. By simulating these dynamics, this research investigates the crucial role of human adaptability in providing systemic resilience and evaluates the overall reliability of HRC under conditions of extreme operational volatility and stochastic robot failures.

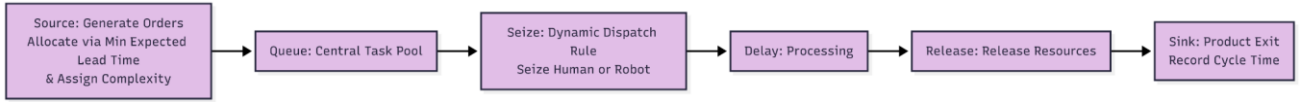


Figure 1. Discrete Event Simulation (DES) flowchart for the order processing logic. Upon generation at the Source, each order is assigned a complexity attribute and placed into a central Queue. A dynamic dispatching rule at the Seize block then allocates the order to either human or robotic resources, optimizing for the minimum expected wait time. Following processing (Delay), resources are freed (Release), and the order exits the system (Sink) for cycle time logging.

4.1 Macro-Process Logic and Scheduling Algorithms

The logistics system and order processing mechanisms of the production line are modeled using Discrete Event Simulation (DES). Upon generation, each order is assigned a continuous complexity attribute, denoted as $C_{order} \in [0,1]$. This parameter quantifies the inherent execution difficulty of specific orders and models its proportional impact on the cognitive failure rate of the humanoid robots.

Regarding order allocation, this study departs from conventional fixed-threshold methodologies. Instead, this study implements a dynamic scheduling logic predicated on the "Minimization of Expected Lead Time." [11] The system continuously evaluates the saturation levels of heterogeneous resources in real time. The expected waiting times for robots ($E[T]_{robot}$) and humans ($E[T]_{human}$) are calculated using the following decision criteria:

$$E[T]_{robot} = \frac{Q_{robot} \times P_{robot}}{N_{robot \text{ active}}} \quad (1)$$

$$E[T]_{human} = \frac{Q_{human} \times P_{human}}{N_{human \text{ active}}} \quad (2)$$

where Q denotes the length of the pending task queue, P represents the nominal processing rate, and N_{active} indicates the current number of available resources for the respective agent type.

Critically, in the event of a robot cognitive failure, the system immediately triggers a highest-priority preemptive signal. This mechanism forces available human operators to interrupt their routine tasks and intervene for immediate fault resolution, thereby maintaining system continuity and reflecting the realistic dynamics of human-robot collaboration.

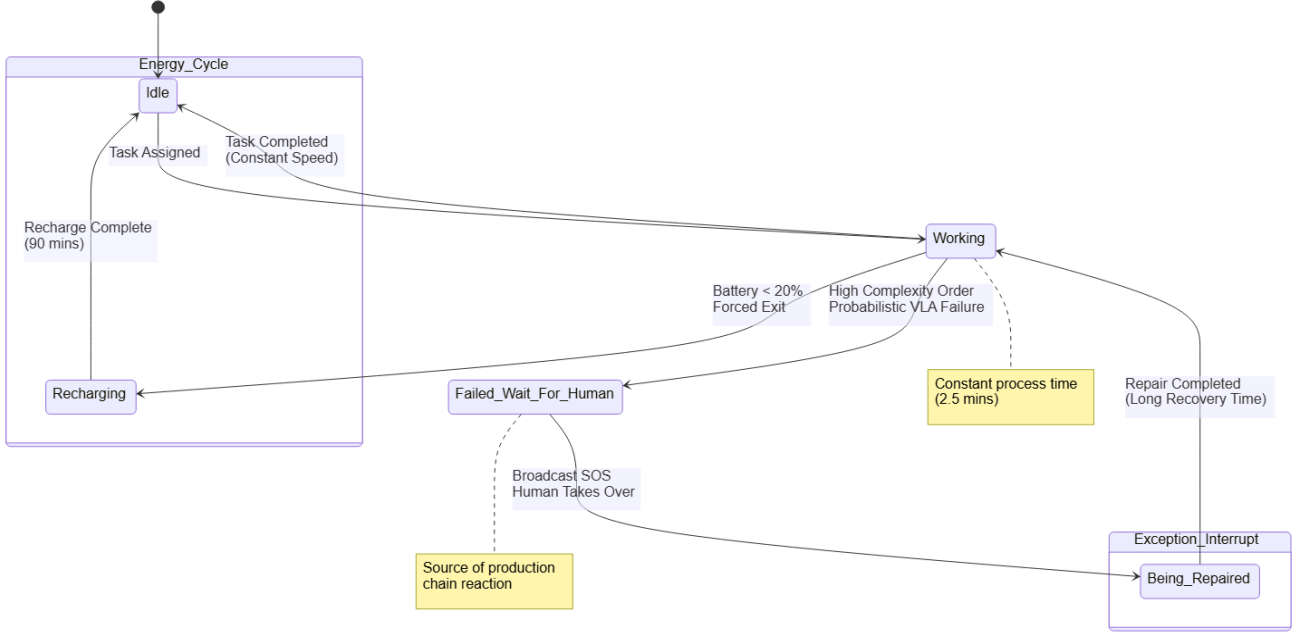


Figure 2 Humanoid Robot Agent State Diagram. The core behavioral logic comprises three primary loops: the Energy Cycle, Routine Working, and Exception Interrupts. Upon experiencing a VLA cognitive failure during high-complexity tasks, the agent enters the (Failed_Wait_For_Human) state. Resuming operations strictly necessitates human operators to interrupt their routine tasks for anomaly resolution (Being_Repaired).

4.2 Agent Behavioral Models and State Transitions

4.2.1 Humanoid Robot Agents

Drawing upon current industry data [6-10], this study establishes empirical parameter baselines for the locomotion and inferential power consumption of humanoid robots. Within the Agent-Based Modeling (ABM) framework, the state transition mechanism of the robot agent is primarily driven by two core constraints: energy cycles and cognitive boundaries. To accurately reflect the high energy consumption inherent in humanoid locomotion and computational reasoning, this model implements a strict energy cycling protocol.

Energy Cycling Protocol: The maximum continuous operational duration for a robot is capped at 240 minutes. While in the (Working) state, the agent continuously depletes its energy reserves. Upon reaching a critical threshold of 20% remaining battery capacity, the robot is forced to exit the production line and transition into a (Recharging) state. This state mandates a 90-minute fast-charging cycle, after which the robot is permitted to return to the production line on standby.

Dynamic Failure Function: To address cognitive boundaries, this study formulates a Complexity-based Cognitive Failure function. Acknowledging the significant discrepancy in success rates of VLA models between standard conditions and unseen edge cases, this model eschews the traditional use of a static error constant. Instead, this research introduces a linear failure function derived from empirical RT-2 success rates [5] to dynamically simulate the cognitive limitations inherent in vision-language-action paradigms.

The probability of cognitive failure (P_f) is defined mathematically as follows:

$$P_f = 0.08 + 0.58 \times C_{order} \quad (3)$$

where C_{order} represents the continuous complexity attribute of the assigned task.

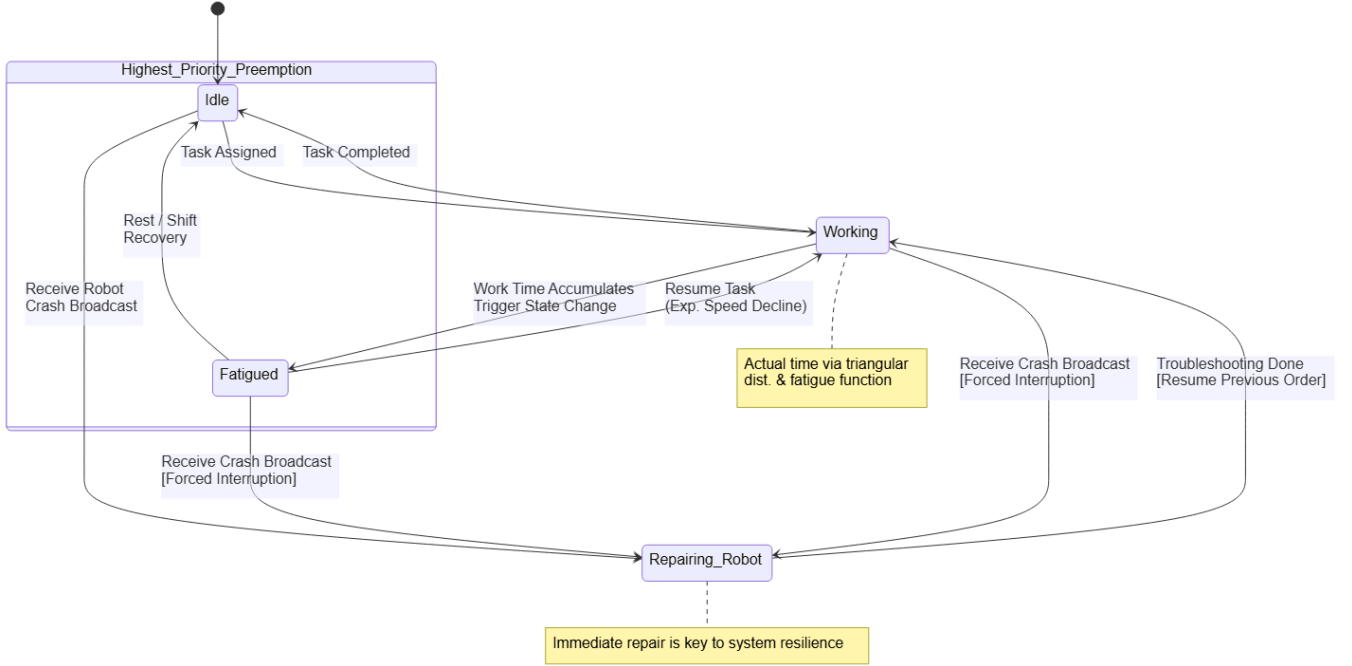


Figure 3 Human Worker Agent State. Diagram. The behavioral logic incorporates routine operations, fatigue accumulation, and a highest-priority preemption mechanism. Upon receiving a robotic failure broadcast, the system forcibly interrupts the operator's current state to prioritize anomaly resolution (Repairing_Robot). This preemptive intervention is essential for maintaining systemic resilience, after which the operator resumes their suspended task.

4.2.2. Human Worker Agents

The configuration of the human model is based on established human fatigue and recovery frameworks [4]. Unlike robotic agents, human worker agents are characterized by cumulative fatigue and high adaptability in anomaly resolution. To account for the inherent stochasticity of individual performance—as human operators do not maintain the uniform execution times characteristic of robots—task durations are modeled using a triangular distribution.

The actual time required for task completion, T_{actual} , is dynamically adjusted by the fatigue coefficient, $V_{fatigue}$, which accumulates over the duration of the work shift. Furthermore, to prevent "burnout" or physical exhaustion, the model incorporates a physiological boundary mechanism. When the cumulative work intensity and duration approach the Maximum Endurance Time (MET), a mandatory rest cycle is triggered, simulating real-world occupational health protocols. The dynamic fatigue and operational duration are mathematically represented as follows:

$$T_{actual} = T_{triangular} \times (1 + V_{fatigue} \times 0.5) \quad (4)$$

4.3 Verification and Validation (V&V)

Model Verification. Model Verification was conducted using Extreme Condition Tests to ensure the programmatic integrity of the simulation logic. In this phase, the robot's cognitive failure probability (P_f) and the human worker's fatigue accumulation rate (λ) were both set to zero, and battery autonomy constraints were removed. Under these idealized conditions, the system should behave as a deterministic discrete-event scheduling model. By observing whether the production output aligns

perfectly with theoretical mathematical predictions, this study verified the correctness of the underlying agent state transitions and entity-routing logic.

Model Validation. Model Validation focused on evaluating the efficacy of the dynamic decision-making and routing mechanisms. Unlike conventional fixed-threshold dispatching, this study implements a heuristic scheduling rule based on the Minimization of Expected Lead Time. [11] Validation was achieved by observing the system's ability to autonomously redistribute task flows to lower-load units under varying resource configurations. This emergent load-balancing behavior confirms that the model accurately captures the complex interactions between heterogeneous resources—humans and robots—within a high-frequency failure environment.

5. Experimental Design & Scenarios

5.1 Experimental Design and Statistical Framework

This research utilizes a full factorial design to construct the simulation framework, aimed at systematically evaluating the interaction effects of varied parameter combinations on the resilience of hybrid human-robot production systems. The primary independent variables are defined as follows: the humanoid robot allocation ratio ($R_{humanoid}$) ranges from 0.1 to 1.0, utilizing a step increment of 0.1 for rigorous stress testing. Simultaneously, task complexity (C_{order}) is categorized into low, medium, and high failure probability (P_f) scenarios based on the cognitive boundaries of VLA models. These correspond to routine tasks ($P_f = 8\%$), general edge cases ($P_f = 37\%$), and environments characterized by extreme uncertainty ($P_f = 66\%$).

To ensure statistical convergence and methodological rigor, this study implements a Monte Carlo simulation framework. For each of the 10×3 parameter combinations, 30 independent iterations were performed. This sample size is consistent with the requirements of the Central Limit Theorem (CLT), ensuring that the sample mean distribution approximates normality and resulting in a total of 900 independent simulation samples. All performance metrics are processed through mean value calculations and supplemented by scatter distribution observations. This approach allows for a granular assessment of system phase transitions and dynamic stability under stochastic perturbations.

5.2 Simulation Scenario Design

To quantify the shift from efficiency-driven optimization to systemic resilience erosion, this study establishes three simulation scenarios. **Scenario A (Baseline Performance)** establishes theoretical peak capacity in highly standardized "Seen Tasks" under low-failure conditions. **Scenario B (Heterogeneous Resource Interaction)** introduces moderate failure pressure, tracking the migration of systemic bottlenecks toward human maintenance workloads as automation increases. Finally, **Scenario C (Stability Boundary Analysis)** simulates and analyzes systemic instability under high task complexity and high robot-to-human ratios to quantify operational volatility. This study posits that severe oscillations and a characteristic bimodal distribution indicate a "resilience trap." This framework provides a realistic assessment of operational unfeasibility, defining the critical physical boundaries required for system steady-state.

5.3 Performance and Stability Evaluation Metrics

To provide a comprehensive assessment of hybrid production line performance, this study constructs a multi-dimensional evaluation framework integrating efficiency, resource dynamics, and risk indicators. In the dimension of productivity efficiency, Total Throughput (TP) and Average Cycle

Time (CT) are employed to measure output scale and systemic flow performance, respectively. Regarding resource status, this research introduces the Bottleneck Offset Index (BI), defined as follows:

$$BI = \frac{Q_{robot} - Q_{human}}{TotalQ} \quad (5)$$

where Q represents the queue length of pending tasks for a specific resource type. A positive BI value indicates that the systemic bottleneck is localized at the robotic units, whereas a negative value signifies a migration of the bottleneck toward the saturation of human maintenance capacity. Finally, the concentration of data points is observed through a visualization analysis of the sample scatter density. A high degree of sample dispersion is interpreted as critical evidence of resilience loss. Such dispersion indicates that the production line has become hyper-sensitive to stochastic cognitive failures and lacks the self-healing capability required to mitigate unpredictable edge cases.

5.4. System Parameters and Variables

Table 1. Global & Environment parameters

Variable Name	Type	Description	Default / Range
p_TotalAgents	int	Total number of task entities generated per simulation run.	50,000 units
p_OrderArrivalRate	double	Order arrival frequency (based on a 24-hour production cycle).	34.72 units/min
p_HumanoidRatio	double	The proportion of humanoid robots in the resource pool	0.0 – 1.0 (Step: 0.1)
p_OrderComplexity	double	Probability of an order being an "Edge Case," triggering VLA cognitive failure.	0.0 ~ 1.0

Table 2. Humanoid Agent parameters

Variable Name	Type	Description	Default / Range
p_RobotBaseTime	double	Deterministic processing time per unit for robotic agents	2.5 min (Fixed)
f_VLA_ExceptionRate()	Function	Probability of cognitive failure based on task complexity	$P_f = 0.08 + 0.58 \times C_{order}$
v_BatteryLevel	double	Current state of charge (SoC) percentage.	0% ~ 100% The battery level drops from over time recharge is triggered when it falls below 20%.
p_BatteryCapacity	double	Maximum continuous operational duration under full load.	240 min
p_RechargeTime	double	Duration of the autonomous fast-charging cycle.	90 min

Table 3.Human Agent parameters

Variable Name	Type	Description	Default / Range
p_HumanTimeDist	double	Baseline processing time utilizing a triangular distribution.	triangular(240,300, 420)s
p_RepairRobotTime	double	Anomaly resolution duration (modeled with a long-tail distribution).	lognormal(1.59, 0.2, 0)分
p_FatigueDecayK	double	Linear fatigue accumulation rate per minute of active work	0.005(Fatigue accumulation rate per minute)
v_FatigueLevel	double	Current cumulative fatigue state (normalized).	$1 - e^{-kt}$
p_MET	double	Maximum Endurance Time; the critical physiological threshold.	Preset for 75 minutes (at a load intensity of 0.3)mentioned in the fatigue model
p_RecoveryRate	double	Recovery Allowance (RA); used to calculate mandatory rest periods.	The preset value is 0.5 (i.e.,50% of the working time) mentioned in the fatigue model
f_ActualProcessTime()	Function	Dynamic processing time adjusted for fatigue degradation.	$T_{actual} = T_{triangular} \times (1 + V_{fatigue} \times 0.5)$

6.Results and Discussion

Through 900 independent simulation samples, this study quantitatively analyzed the dynamic performance of the human-robot configuration ratio ($p_HumanoidRatio$) under various task complexity scenarios. The discrete distribution visualized via Python (Figure 4) reveals that systemic performance does not exhibit a linear correlation with increased automation levels. Instead, the results indicate that system efficiency is significantly constrained by two primary bottlenecks: the saturation of maintenance capacity and the physiological boundaries of human operators.

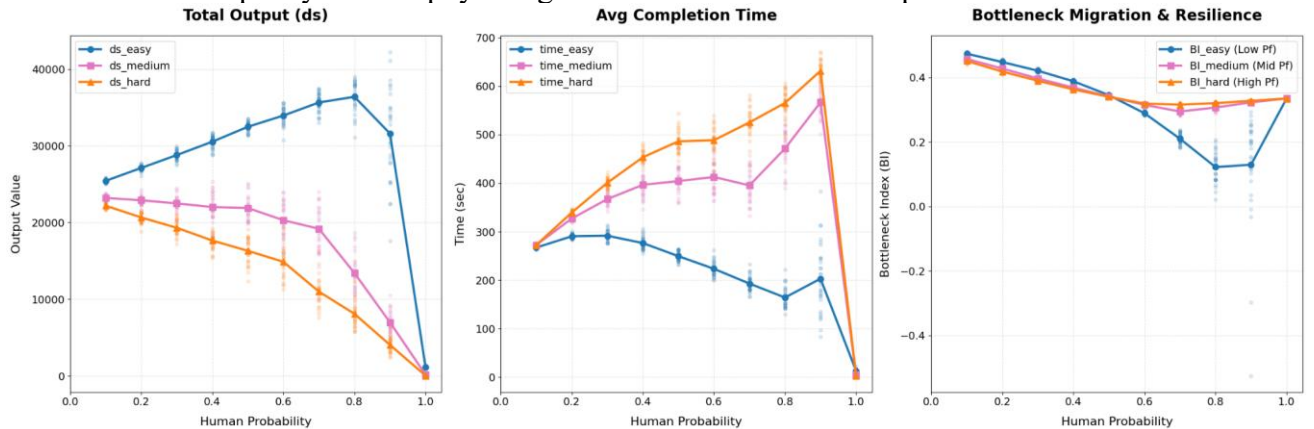


Figure 4. Multi-dimensional performance and resilience analysis of human-robot allocation ratios across varying task complexities.

6.1 Analysis of Critical Tipping Points in Throughput and Stability

The total throughput distribution graph is evidence that as you increase the ratio of humanoid robots, it results in a non-linear yielding of maximum throughput by the system, even for low-

complexity test case "ds_easy". The empirical results indicate that the maximum capacity output occurs at a human-robot ratio of 0.8 (20% human) and will generate, on average, approximately 37,000 units per period. However, while the maximum mean load occurs at the 0.8 ratio, the variation in the sample load produced at the 0.8 ratios is significantly greater than the same measure of variation at the 0.7 ratios. This pattern indicates that failures that occur randomly are contributing to the depletion of maintenance resources and thus the overall stability of the system as a whole is declining.

The humanoid ratio reaches 0.9, making the system experience the highest levels of unpredictable and volatile operations. Majority of the data points measured at this level have bimodal distributions; only few random-sequences had high-performance-level because they had no long tail failures while most-sequences produce less than 20,000 units since they enter actual-deadlock situations. This confirms the basic hypothesis of this research, as once all maintenance resources in highly automated environments are utilized, there is an immediate shift from "high-efficiency" operating conditions to being-completely-paralyzed structurally. Therefore, maintaining a humanoid ratio of 0.9 or above creates an unstable and risky manufacturing environment.

6.2 Correlation between Completion Time and Human Performance Degradation

Analysis of average order processing times reveals divergent trends depending on task complexity. In *easy case* scenarios, completion times exhibit a consistent downward trend, demonstrating that humanoid robots effectively accelerate production in low-failure environments. Conversely, in *medium* and *hard case* scenarios, the high frequency of cognitive failures significantly escalates maintenance time costs. The efficiency gains traditionally associated with robotic processing speed are entirely offset by the penalties of high failure rates, leading to an overall upward trend in lead times.

These results indicate that in environments characterized by high logical failure rates, human-robot collaboration imposes a significant operational overhead. As the robot-to-human ratio increases, the average order completion time follows a monotonically increasing trend. This phenomenon can be elucidated by the Jaber & Neumann fatigue model: a higher density of deployed robots precipitates a greater frequency of interruptive maintenance requests, which perpetually disrupts human operators during routine processing tasks. Such high-frequency task switching accelerates fatigue accumulation. Furthermore, due to the "long-tail" nature of maintenance interventions, these disruptions induce cascading delays within the central task pool.

The exponential increase in completion times at high-automation configurations further reflects the exhaustion of human resources. As robot failure rates climb, human maintenance capacity rapidly approaches the physiological boundary of Maximum Endurance Time (MET), ultimately triggering a catastrophic collapse or "avalanche" in system throughput.

6.3 Bottleneck Migration and the Quantification of Resilience Value

The successful quantification of the bottleneck migration process in this study is evidenced by the discrete fluctuations in the Work-In-Process (WIP) index. Lower humanoid ratios are primarily correlated with processing bottlenecks at robotic units, manifested as positive WIP index values. However, as the humanoid ratio increases, the WIP curve converges toward the zero-axis and drops precipitously at a ratio of 0.9 (reaching a value of -0.5). This transition indicates a fundamental shift in the systemic bottleneck: the constraint migrates from production throughput to human maintenance availability.

The empirical data suggests that a humanoid ratio of 0.7 (representing a 30% human worker buffer) yields the optimal resilient stability. The highly concentrated sample distribution at this configuration demonstrates that sufficient human resource buffers are capable of effectively offsetting the stochastic pressure induced by VLA cognitive failures. This finding quantifies the value of human presence as a critical resilience factor in maintaining system stability amidst the inherent uncertainties of embodied AI.

7. Conclusion and Future Work

Departing from conventional literature that assumes constant reliability for automated equipment, this research is among the first to incorporate the cognitive failure probabilities of VLA models in edge cases and high-load energy consumption (fast-charging cycles) as systemic constraints. Furthermore, by integrating human worker models encompassing dynamic fatigue and Maximum Endurance Time (MET), this study quantitatively evaluate the systemic resilience boundaries under extreme demand volatility.

Monte Carlo simulations reveal a non-linear relationship between total throughput and the automation ratio. While maximizing the robot ratio benefits standardized tasks, system performance in high-complexity environments is strictly governed by maintenance resource saturation and human fatigue. Notably, exceeding an 80% humanoid ratio (i.e., human personnel below 20%) triggers a severe "Resilience Trap." This leads to scheduling deadlocks and capacity collapse as bottlenecks irreversibly migrate to human maintenance resources. Therefore, maintaining a human worker ratio of 20% to 30% is not merely an operational baseline, but an indispensable resilience buffer to absorb stochastic pressures until AI logical hallucinations and power density limits are fully overcome.

While this study quantifies dynamic losses in HRC, it currently excludes physical factory layouts and motion-path interference. Future research will extend this framework to explore Multi-Agent Systems (MAS) for robot-to-robot repair protocols and on-device learning mechanisms. By simulating the dynamic reduction of robot failure rates along a learning curve, future models can more accurately predict the long-term resource evolution of next-generation intelligent manufacturing systems.

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